



**Java OOPs:
Inventory Management System
Assignment - 4**

Problem Statement:

You have been hired to design and develop an Inventory Management System for a small retail store. The store needs a comprehensive solution to efficiently track its inventory, handle sales, and manage product information.

Requirements:

1. **Product Class:** Create a Product class to represent individual products in the store. Each product should have the following attributes:
 - a. Product ID
 - b. Product Name
 - c. Category (e.g., electronics, clothing, groceries)
 - d. Unit Price
 - e. Quantity in Stock
2. **Inventory Class:** Create an Inventory class that will manage the store's product inventory. The Inventory class should have methods for the following functionalities:
 - a. Adding a new product to the inventory.
 - b. Updating product information (e.g., price, quantity) for existing products.
 - c. Removing products from the inventory.
 - d. Displaying the list of all products in stock.
 - e. Searching for products by name or category.
 - f. Generating a report of low stock items (products with quantity below a specified threshold).
3. **Sales Class:** Create a Sales class to handle sales transactions. The Sales class should have methods for the following functionalities:
 - a. Adding items to the shopping cart.
 - b. Calculating the total bill for the customer.
 - c. Reducing the quantity of sold items from the inventory after a successful sale.
4. **User Interface:** Implement a user interface in a separate Main class to interact with the Inventory and Sales classes. The user interface should:
 - a. Display a menu with options to manage inventory and handle sales.

- b. Allow the user to add, update, or remove products from the inventory.
 - c. Provide options for searching products by name or category.
 - d. Allow the user to add items to the shopping cart and calculate the total bill.
5. **Error Handling:** Implement error handling for potential input-related exceptions, such as invalid product IDs or quantities during sales transactions.